

Yue Ying

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EDUCATION

Aug 2021 - Present **Johns Hopkins University, Whiting School of Engineering**, Baltimore, MD, USA
PhD, Chemical & Biochemical Engineering

Aug 2017 - May 2021 **Franklin & Marshall College**, Lancaster, PA, USA
BA, Biochemistry and Molecular Biology & German Language and Culture Major; Theoretical Math Minor, (GPA 3.84 / 4.00)

Sep 2019 - Dec 2019 **Ruprecht-Karls-Universität Heidelberg**, Heidelberg, Germany
Non-degree, American Junior Year (AJY) at Heidelberg Exchange Program German Studies, (German Grade 1.5)

RESEARCH EXPERIENCE

Sep 2021 - Present *Researcher*
Dr. Margaret Johnson Lab, Jenkins Dpt. of Biophysics, Johns Hopkins University

- Investigated influence of membrane bending energy selects for growth pathway of protein assemblies
- Implemented continuum membrane kinetics in Python applying Brownian dynamics of Helfrich Hamiltonian membrane model
- Developed a script that takes in PDB code and automatically builds a Coarse-grained model, creates a corresponding reaction system, solves for ODE solution, and runs Gillespie simulation

May 2020 - Aug 2021 *Researcher*
Dr. Deanne Taylor Lab, Children's Hospital of Philadelphia, University of Pennsylvania
TRIP (Translational Research Internship Program at University of Pennsylvania)
COV-IRT (COVID-19 International Research Team, online)

- Built metabolic models and conducted flux balance analysis of NASA gene lab space mice model using Cobra and CORDA in Python
- Visualized the analysis results using reaction heat map created with Escher, which eases interpretation of data for researchers with a clinical background

Nov 2018 - May 2020 *Researcher*
Dr. Jason W. Labonte Lab, Franklin & Marshall College / Johns Hopkins University
Rosetta Commons (rosettacommons.org, online)

- Conducted research projects on combinatorial expansion of the carbohydrate database of the Rosetta software
- Added molecule parameter files of over 800 different monosaccharide residues to satisfy the need of unusual monosaccharide residues such as in antibody design

INTERNSHIP & TEACHING EXPERIENCE

Sep 2022 - Dec 2023 *Teaching Assistant for the course "Modeling the Living Cells"*

Johns Hopkins University, Baltimore, MD, USA

- Tutored undergraduate students
- Designed and lead computational lab sessions

Jan 2020 - May 2020 *Organic Chemistry Lab Assistant*

Franklin & Marshall College, Lancaster, PA, USA

- Prepared and facilitated teaching labs and tutored peers

Aug 2018 - May 2019 *New College House Mentor Advisor*

Franklin & Marshall College, Lancaster, PA, USA

- Tutored freshmen ('22) of New College House, Franklin & Marshall College
- Organized group gatherings, study sessions, and volunteering activities

Jun 2017 - Aug 2017 *Software Developer (Internship)*

SIPM Information Technology Co. Ltd., Shanghai, China

- Built the English and the Japanese GUI
- Built the environment for a distributed database
- Developed algorithm for two-sided assembly line balancing with Java, which is integrated into the algorithm package provided by the company to its clients

Jul 2016 - Aug 2016 *Software Developer (Internship)*

CIMC (China International Marine Containers) Co. Ltd., Jiangsu, China

- Developed scheduling optimization algorithm with Java, which shows that the scheduling and assembly line arrangement had a great potential of optimization

PUBLICATIONS

Ying, Y.*, Sang, M.*, Au, G., Chhibber, S., Du, Y., Fischer, J.A., Foley, S.L., Guo, S., Herzog-Pohl, I., Liu, Z., Roscom, H., Sohail, H., Sho, S.S., Johnson, M.E. (2026) Transforming macromolecular structures into simulations of self-assembly with ioNERDSS. *Biophys. J.* (in revision)
<https://doi.org/10.64898/2026.01.27.702082> (* co-first author)

Ying, Y., Johnson, M.E. (2026) Membrane bending energy selects for compact growth of protein assemblies. *Phys. Rev. Res.* (accepted) <https://doi.org/10.1103/tnv7-fjh5>

Fischer, J.A., Greenberg, E., **Ying, Y.**, Sho, S.S., Foley, S.L., Johnson, M.E. (2025) A membrane-driven biochemical oscillator tunable by the volume to surface area ratio. *BioRxiv*.
<https://doi.org/10.1101/2025.11.14.688573>

Guo, S., Korolija, N., Milfeld, K., Jhaveri, A., Sang, M., **Ying, Y.**, Johnson, M.E. (2025) Parallelization of particle-based reaction diffusion simulations using MPI. *J. Comput. Chem.*, 46, e70132.
<https://doi.org/10.1002/jcc.70132>

da Silveira, W.A., Fazelinia, H., Rosenthal, S.B., Laiakis, E.C., Kim, M.S., Meydan, C., Kidane, Y., Rathi, K.S., Smith, S.M., Stear, B., **Ying, Y.**, Zhang, Y., ... Beheshti, A. (2020). Comprehensive Multi-omics Analysis Reveals Mitochondrial Stress as a Central Biological Hub for Spaceflight Impact. *Cell* 183, 1185-1201.e20. <https://doi.org/10.1016/j.cell.2020.11.002>

TALKS & PRESENTATIONS

Ying, Y. (2026, Feb). "Transforming PDB structures into coarse-grained reaction-diffusion simulations with ioNERDSS." Internal talk at JHU.

Ying, Y. (2025, Mar). "Membrane bending energy selects for compact growth of protein assemblies." Paper presented at APS Global Physics Summit 2025, Anaheim, CA.

Ying, Y. (2024, Nov). "Protein assemblies prefer regular growth due to membrane energy penalty during budding." Presented at JHU Soft matter and biological physics group meeting.

Ying, Y. (2024, Feb). "A dynamic and mechanical membrane model to study membrane remodeling kinetics." Presented at BPS Annual Meeting 2024, Philadelphia, PA.

Ying, Y. & Fu, Y. (2023, Mar). "A dynamic and mechanical membrane model to study membrane remodeling kinetics." Presented at APS March Meeting 2023, Las Vegas, NV.

HONORS & AWARDS

Mar 2020 *Member*
Delta Phi Alpha National German Honor Society, Tau Gamma Chapter

Jan 2015 *Outstanding*
High School Mathematical Contest in Modeling (HiMCM) 2014, COMAP

HOBBIES & INTERESTS

Sep 2017 - May 2021 *Violin I / Violin II*
F&M Orchestra and Philharmonia, Franklin & Marshall College

OTHER SKILLS

IT-Skills Python | C/C++ | Java | Lua | Linux bash | LaTeX | Github | PyMOL

Languages Mandarin Chinese (native) | English (fluent, US undergraduate)
German (fluent, DSH-2, equivalent to CEFR C1)
Japanese (medium)

Baltimore, MD, Jun. 27, 2026